

Lecture Notes: Definite Integrals

Calculus I

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Section 1 Area and Estimating with Finite Sums

Definition 1 (Area Problem) Given a function $f(x) \geq 0$ on an interval $[a, b]$, we wish to find the area A of the region bounded by the graph of f , the x -axis, and the vertical lines $x = a$ and $x = b$.

Definition 2 (Partition) A partition of $[a, b]$ is a finite set of points $P = \{x_0, x_1, x_2, \dots, x_n\}$ such that

$$a = x_0 < x_1 < x_2 < \dots < x_n = b.$$

The subintervals are $[x_{i-1}, x_i]$ for $i = 1, 2, \dots, n$. The width of the i -th subinterval is $\Delta x_i = x_i - x_{i-1}$. If all subintervals have the same width, then $\Delta x = \frac{b-a}{n}$.

Definition 3 (Riemann Sum) Let f be defined on $[a, b]$ and let P be a partition of $[a, b]$. Let c_i be any sample point in the subinterval $[x_{i-1}, x_i]$. The sum

$$S = \sum_{i=1}^n f(c_i) \Delta x_i$$

is called a Riemann sum for f on $[a, b]$.

Example 1 Estimate the area under $f(x) = x^2$ on $[0, 1]$ using four rectangles and right endpoints. With $n = 4$, $\Delta x = \frac{1-0}{4} = 0.25$. The right endpoints are $c_1 = 0.25$, $c_2 = 0.5$, $c_3 = 0.75$, $c_4 = 1.0$.

$$\begin{aligned} S_4 &= f(0.25) \cdot 0.25 + f(0.5) \cdot 0.25 + f(0.75) \cdot 0.25 + f(1.0) \cdot 0.25 \\ &= (0.0625 + 0.25 + 0.5625 + 1) \cdot 0.25 = 1.875 \cdot 0.25 = 0.46875. \end{aligned}$$

Using left endpoints: $c_1 = 0$, $c_2 = 0.25$, $c_3 = 0.5$, $c_4 = 0.75$.

$$S_4 = (0 + 0.0625 + 0.25 + 0.5625) \cdot 0.25 = 0.875 \cdot 0.25 = 0.21875.$$

The actual area is $\frac{1}{3} \approx 0.3333$.

Definition 4 (Lower and Upper Sums) If f is bounded on $[a, b]$, the lower sum $L(f, P)$ uses the minimum value of f on each subinterval, and the upper sum $U(f, P)$ uses the maximum value.

Section 2 Sigma Notation and Limits of Finite Sums

Definition 5 (Sigma Notation) The sum $a_1 + a_2 + \dots + a_n$ is written compactly as

$$\sum_{i=1}^n a_i,$$

where i is the index of summation, 1 is the lower limit, and n is the upper limit.

Theorem 1 (Properties of Sigma Notation) 1. $\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$.

2. $\sum_{i=1}^n c a_i = c \sum_{i=1}^n a_i$ for any constant c .

3. $\sum_{i=1}^n c = nc$.

$$4. \sum_{i=1}^n i = \frac{n(n+1)}{2}.$$

$$5. \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$$

$$6. \sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

Example 2 Evaluate $\sum_{i=1}^{100} (2i - 1)$.

$$\sum_{i=1}^{100} (2i - 1) = 2 \sum_{i=1}^{100} i - \sum_{i=1}^{100} 1 = 2 \cdot \frac{100 \cdot 101}{2} - 100 = 10100 - 100 = 10000.$$

Definition 6 (Limit of a Riemann Sum) Let f be defined on $[a, b]$. The definite integral of f from a to b is

$$\int_a^b f(x) dx = \lim_{\|P\| \rightarrow 0} \sum_{i=1}^n f(c_i) \Delta x_i,$$

provided this limit exists and is the same for all partitions P and all choices of sample points c_i . Here $\|P\|$ is the norm of the partition, the largest subinterval width.

Theorem 2 (Integrability) If f is continuous on $[a, b]$, or if f is bounded on $[a, b]$ and has only finitely many discontinuities, then f is integrable on $[a, b]$.

Example 3 Evaluate $\int_0^1 x^2 dx$ using the limit definition. Divide $[0, 1]$ into n equal subintervals. Then $\Delta x = \frac{1}{n}$ and $c_i = \frac{i}{n}$ (right endpoints).

$$S_n = \sum_{i=1}^n f\left(\frac{i}{n}\right) \frac{1}{n} = \sum_{i=1}^n \left(\frac{i}{n}\right)^2 \frac{1}{n} = \frac{1}{n^3} \sum_{i=1}^n i^2 = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2}.$$

Taking the limit as $n \rightarrow \infty$,

$$\int_0^1 x^2 dx = \lim_{n \rightarrow \infty} \frac{(n+1)(2n+1)}{6n^2} = \lim_{n \rightarrow \infty} \frac{2n^2 + 3n + 1}{6n^2} = \frac{2}{6} = \frac{1}{3}.$$

Section 3 The Definite Integral

Definition 7 (Definite Integral) The definite integral of a function f from a to b is the number

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x,$$

where $\Delta x = \frac{b-a}{n}$ and c_i is any point in the i -th subinterval $[x_{i-1}, x_i]$.

Theorem 3 (Properties of Definite Integrals) 1. $\int_a^a f(x) dx = 0$.

$$2. \int_a^b f(x) dx = - \int_b^a f(x) dx.$$

$$3. \int_a^b c \, dx = c(b - a).$$

$$4. \int_a^b [f(x) + g(x)] \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx.$$

$$5. \int_a^b c f(x) \, dx = c \int_a^b f(x) \, dx.$$

$$6. \int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx \text{ for any } c \in [a, b].$$

$$7. \text{ If } f(x) \geq 0 \text{ on } [a, b], \text{ then } \int_a^b f(x) \, dx \geq 0.$$

$$8. \text{ If } f(x) \geq g(x) \text{ on } [a, b], \text{ then } \int_a^b f(x) \, dx \geq \int_a^b g(x) \, dx.$$

Example 4 Evaluate $\int_0^2 (3x^2 - 2x) \, dx$ using the limit definition. $\Delta x = \frac{2}{n}$, $c_i = \frac{2i}{n}$.

$$S_n = \sum_{i=1}^n \left(3 \left(\frac{2i}{n} \right)^2 - 2 \left(\frac{2i}{n} \right) \right) \frac{2}{n} = \sum_{i=1}^n \left(\frac{12i^2}{n^2} - \frac{4i}{n} \right) \frac{2}{n} = \frac{24}{n^3} \sum_{i=1}^n i^2 - \frac{8}{n^2} \sum_{i=1}^n i.$$

Using the sum formulas,

$$S_n = \frac{24}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} - \frac{8}{n^2} \cdot \frac{n(n+1)}{2} = \frac{4(n+1)(2n+1)}{n^2} - \frac{4(n+1)}{n}.$$

Taking the limit as $n \rightarrow \infty$,

$$\int_0^2 (3x^2 - 2x) \, dx = 8 - 4 = 4.$$

Section 4 The Fundamental Theorem of Calculus

Theorem 4 (Fundamental Theorem of Calculus, Part 1) If f is continuous on $[a, b]$, then the function F defined by

$$F(x) = \int_a^x f(t) \, dt, \quad a \leq x \leq b,$$

is continuous on $[a, b]$, differentiable on (a, b) , and $F'(x) = f(x)$ for all $x \in (a, b)$.

Let $x \in (a, b)$ and $h \neq 0$ such that $x + h \in (a, b)$. Then

$$F(x+h) - F(x) = \int_a^{x+h} f(t) \, dt - \int_a^x f(t) \, dt = \int_x^{x+h} f(t) \, dt.$$

By the Mean Value Theorem for Integrals, there exists c between x and $x+h$ such that

$$\int_x^{x+h} f(t) \, dt = f(c)h.$$

Thus, $\frac{F(x+h)-F(x)}{h} = f(c)$. As $h \rightarrow 0$, $c \rightarrow x$, and since f is continuous, $f(c) \rightarrow f(x)$. Hence $F'(x) = f(x)$.

Theorem 5 (Fundamental Theorem of Calculus, Part 2) *If f is continuous on $[a, b]$ and F is any antiderivative of f on $[a, b]$, then*

$$\int_a^b f(x) dx = F(b) - F(a).$$

Let $G(x) = \int_a^x f(t) dt$. By Part 1, $G'(x) = f(x)$, so G is an antiderivative of f . Since F is also an antiderivative, $F(x) = G(x) + C$ for some constant C . Then

$$F(b) - F(a) = (G(b) + C) - (G(a) + C) = G(b) - G(a) = \int_a^b f(t) dt - \int_a^a f(t) dt = \int_a^b f(t) dt.$$

Remark 1 *The notation $F(x) \Big|_a^b = F(b) - F(a)$ is often used.*

Example 5 *Evaluate $\int_1^3 x^2 dx$. An antiderivative of x^2 is $\frac{x^3}{3}$. Thus,*

$$\int_1^3 x^2 dx = \frac{x^3}{3} \Big|_1^3 = \frac{27}{3} - \frac{1}{3} = \frac{26}{3}.$$

Example 6 *Evaluate $\int_0^{\pi/2} \cos x dx$. An antiderivative of $\cos x$ is $\sin x$. Thus,*

$$\int_0^{\pi/2} \cos x dx = \sin x \Big|_0^{\pi/2} = \sin \frac{\pi}{2} - \sin 0 = 1 - 0 = 1.$$

Example 7 *Evaluate $\int_1^e \frac{1}{x} dx$. An antiderivative of $\frac{1}{x}$ is $\ln|x|$. Thus,*

$$\int_1^e \frac{1}{x} dx = \ln x \Big|_1^e = \ln e - \ln 1 = 1 - 0 = 1.$$

Theorem 6 (Net Change Theorem) *The integral of a rate of change is the net change:*

$$\int_a^b F'(x) dx = F(b) - F(a).$$

Example 8 *A particle moves along a line with velocity $v(t) = t^2 - 4t + 3$ m/s. Find the displacement from $t = 0$ to $t = 4$.*

$$\text{Displacement} = \int_0^4 (t^2 - 4t + 3) dt = \left[\frac{t^3}{3} - 2t^2 + 3t \right]_0^4 = \left(\frac{64}{3} - 32 + 12 \right) - 0 = \frac{64}{3} - 20 = \frac{4}{3} \text{ m}.$$

Section 5 Substitution in Definite Integrals

Theorem 7 (Substitution Rule for Definite Integrals) *If g' is continuous on $[a, b]$ and f is continuous on the range of g , then*

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.$$

Let F be an antiderivative of f . Then $\frac{d}{dx}F(g(x)) = F'(g(x))g'(x) = f(g(x))g'(x)$. By the Fundamental Theorem of Calculus,

$$\int_a^b f(g(x))g'(x) dx = F(g(x)) \Big|_a^b = F(g(b)) - F(g(a)).$$

Also, $\int_{g(a)}^{g(b)} f(u) du = F(u) \Big|_{g(a)}^{g(b)} = F(g(b)) - F(g(a))$. The equality follows.

Example 9 Evaluate $\int_0^1 2xe^{x^2} dx$. Let $u = x^2$. Then $du = 2x dx$. When $x = 0$, $u = 0$. When $x = 1$, $u = 1$. Thus,

$$\int_0^1 2xe^{x^2} dx = \int_0^1 e^u du = e^u \Big|_0^1 = e - 1.$$

Example 10 Evaluate $\int_0^{\pi/2} \sin^3 x \cos x dx$. Let $u = \sin x$. Then $du = \cos x dx$. When $x = 0$, $u = 0$. When $x = \pi/2$, $u = 1$. Thus,

$$\int_0^{\pi/2} \sin^3 x \cos x dx = \int_0^1 u^3 du = \frac{u^4}{4} \Big|_0^1 = \frac{1}{4}.$$

Example 11 Evaluate $\int_1^2 \frac{1}{2x+1} dx$. Let $u = 2x+1$. Then $du = 2 dx$, so $dx = \frac{du}{2}$. When $x = 1$, $u = 3$. When $x = 2$, $u = 5$. Thus,

$$\int_1^2 \frac{1}{2x+1} dx = \int_3^5 \frac{1}{u} \cdot \frac{du}{2} = \frac{1}{2} \ln |u| \Big|_3^5 = \frac{1}{2} (\ln 5 - \ln 3) = \frac{1}{2} \ln \frac{5}{3}.$$

Theorem 8 (Integrals of Even and Odd Functions) If f is continuous on $[-a, a]$, then:

1. If f is even, $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$.
2. If f is odd, $\int_{-a}^a f(x) dx = 0$.

Example 12 Evaluate $\int_{-2}^2 (x^3 + x) dx$. The function $f(x) = x^3 + x$ is odd, so the integral is 0.

Example 13 Evaluate $\int_{-1}^1 x^2 dx$. The function $f(x) = x^2$ is even, so

$$\int_{-1}^1 x^2 dx = 2 \int_0^1 x^2 dx = 2 \cdot \frac{1}{3} = \frac{2}{3}.$$

Section 6 Improper Integrals

Definition 8 (Improper Integral of Type I: Infinite Interval) If $\int_a^t f(x) dx$ exists for every $t \geq a$, then

$$\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx,$$

provided the limit exists. Similarly,

$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx,$$

and

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx$$

for any real number c . If the limit exists, the improper integral converges; otherwise, it diverges.

Example 14 Evaluate $\int_1^{\infty} \frac{1}{x^2} dx$.

$$\int_1^{\infty} \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \int_1^t x^{-2} dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{x} \right]_1^t = \lim_{t \rightarrow \infty} \left(-\frac{1}{t} + 1 \right) = 1.$$

The integral converges to 1.

Example 15 Evaluate $\int_1^{\infty} \frac{1}{x} dx$.

$$\int_1^{\infty} \frac{1}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} [\ln x]_1^t = \lim_{t \rightarrow \infty} \ln t = \infty.$$

The integral diverges.

Theorem 9 (p-Integral Test) The improper integral $\int_1^{\infty} \frac{1}{x^p} dx$ converges if $p > 1$ and diverges if $p \leq 1$.

For $p \neq 1$,

$$\int_1^t \frac{1}{x^p} dx = \left[\frac{x^{1-p}}{1-p} \right]_1^t = \frac{t^{1-p} - 1}{1-p}.$$

If $p > 1$, then $1 - p < 0$, so $t^{1-p} \rightarrow 0$ as $t \rightarrow \infty$, giving $\frac{1}{p-1}$. If $p < 1$, then $1 - p > 0$, so $t^{1-p} \rightarrow \infty$, and the integral diverges. For $p = 1$, the integral is $\ln t \rightarrow \infty$.

Definition 9 (Improper Integral of Type II: Infinite Discontinuity) If f is continuous on $[a, b)$ and has an infinite discontinuity at b , then

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx,$$

provided the limit exists. Similarly, if f has an infinite discontinuity at a ,

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx.$$

If f has a discontinuity at $c \in (a, b)$, then

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

Example 16 Evaluate $\int_0^1 \frac{1}{\sqrt{x}} dx$. The integrand has an infinite discontinuity at $x = 0$.

$$\int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{t \rightarrow 0^+} \int_t^1 x^{-1/2} dx = \lim_{t \rightarrow 0^+} [2\sqrt{x}]_t^1 = \lim_{t \rightarrow 0^+} (2 - 2\sqrt{t}) = 2.$$

The integral converges to 2.

Example 17 Evaluate $\int_0^1 \frac{1}{x} dx$. The integrand has an infinite discontinuity at $x = 0$.

$$\int_0^1 \frac{1}{x} dx = \lim_{t \rightarrow 0^+} \int_t^1 \frac{1}{x} dx = \lim_{t \rightarrow 0^+} [\ln x]_t^1 = \lim_{t \rightarrow 0^+} (0 - \ln t) = \infty.$$

The integral diverges.

Theorem 10 (Comparison Test for Improper Integrals) Suppose f and g are continuous on $[a, \infty)$ with $0 \leq f(x) \leq g(x)$ for all $x \geq a$.

1. If $\int_a^\infty g(x) dx$ converges, then $\int_a^\infty f(x) dx$ converges.
2. If $\int_a^\infty f(x) dx$ diverges, then $\int_a^\infty g(x) dx$ diverges.

Example 18 Determine whether $\int_1^\infty \frac{1}{x^3 + 1} dx$ converges. For $x \geq 1$, $x^3 + 1 > x^3$, so $\frac{1}{x^3 + 1} < \frac{1}{x^3}$. Since $\int_1^\infty \frac{1}{x^3} dx$ converges (p -integral with $p = 3 > 1$), by the Comparison Test, the given integral also converges.

Example 19 Determine whether $\int_1^\infty \frac{1}{\sqrt{x} + 1} dx$ converges. For $x \geq 1$, $\sqrt{x} + 1 \leq 2\sqrt{x}$, so $\frac{1}{\sqrt{x} + 1} \geq \frac{1}{2\sqrt{x}}$. Since $\int_1^\infty \frac{1}{2\sqrt{x}} dx = \frac{1}{2} \int_1^\infty x^{-1/2} dx$ diverges (p -integral with $p = 1/2 \leq 1$), by the Comparison Test, the given integral diverges.

Theorem 11 (Absolute Convergence) If $\int_a^\infty |f(x)| dx$ converges, then $\int_a^\infty f(x) dx$ also converges. In this case, we say the integral is absolutely convergent.

Example 20 Determine whether $\int_1^\infty \frac{\sin x}{x^2} dx$ converges. Since $|\frac{\sin x}{x^2}| \leq \frac{1}{x^2}$ and $\int_1^\infty \frac{1}{x^2} dx$ converges, the integral converges absolutely, hence converges.

Remark 2 Improper integrals are essential in probability theory, physics, and engineering, particularly in the study of continuous distributions and Fourier analysis.